Stakeholder Identification, Classification, and Collaborative Mechanisms

A Practice-Based Analysis of the ByeGerm Project

Abstract

This study examines stakeholder identification, classification, and collaborative mechanisms in the ByeGerm portable rapid respiratory-pathogen detection project. Drawing on Freeman's stakeholder theory, Frederick's distinction between direct and indirect stakeholders, and Clarkson's primary-secondary stakeholder classification, the study introduces a three-dimensional analytical framework comprising degree of association, directness of influence, and continuity dependence to identify stakeholder groups and construct a stakeholder map. On this basis, a power-interest matrix is employed to determine management priorities, and differentiated communication and coordination strategies are developed for distinct stakeholder groups. Project practice indicates that the proposed framework can help anchor technological development in genuine societal needs and support dynamic alignment between research and development and real-world application contexts.

Keywords: stakeholders; power-interest matrix; public health

1. Introduction

Portable rapid detection technologies for respiratory pathogens are undergoing a paradigm shift from centralized specialist laboratories toward primary-care and community public-health settings [1]. However, the transition from a research prototype to real-world implementation is not a simple linear process. Different degrees of translational barriers may arise during needs identification, technology development, solution iteration, product manufacturing, contextual adaptation, end-user operation, and public-health application [2]. These barriers do not stem from a single factor; rather, they emerge from the interaction of multiple constraints, including mismatches in primary-care resource allocation, insufficient operational adherence among target populations, fragmented multisource data, and delayed alignment between public policy and regulatory mechanisms. Collectively, these features are characteristic of a complex, nonlinear social system [3].

The BMJ framework for complex interventions emphasizes that the effectiveness of a health intervention depends not only on the intervention itself but also on the implementation context, participating actors, and their interactions. Accordingly, attention should be paid throughout development, evaluation, and implementation to real-world contextual fit, stakeholder needs, and key uncertainties [4]. Research on early health technology assessment further indicates that the experiences and value judgments of patients and the public can complement the limitations of conventional clinical evidence and broaden participation in technology assessment and decision-making [5]. Existing studies also suggest that stakeholder engagement should not be reduced to the simple inclusion of multiple parties; instead, the relevant participants, timing of engagement, and decision-making roles should be clearly defined to establish an effective engagement mechanism [6]. Therefore, for medical technologies spanning research and development, manufacturing, contextual application, and public-health governance, successful translation depends not only on technical performance but also on the needs, roles, and interactions of diverse actors.

Systematically identifying these actors and clarifying their relationships across different stages is essential for understanding the translational process and its collaborative mechanisms. On this basis, the present study applies stakeholder theory to systematically identify, classify, and analyze the relationships among stakeholders involved in the translation of ByeGerm.

Stakeholder theory originated in the field of strategic management in the 1980s. Freeman defined stakeholders as individuals or groups that can affect the achievement of organizational objectives or are affected by the process through which those objectives are achieved [7]. As a macro-level analytical tool for complex systems, stakeholder theory can be used to clarify the flows of power, resources, and value within social networks [8]. By constructing stakeholder maps, researchers can identify critical project nodes, assess the relative weight of stakeholder claims, and seek resource-allocation arrangements that approach Pareto optimality under conditions of competing interests [9].

For the present project, the use of a stakeholder model is particularly important in practice. First, the ByeGerm life cycle spans a highly fragmented network encompassing technology development, manufacturing, primary-level application, and public-health regulation. Without a systematic understanding of diverse actors and their collaborative relationships, technological development may become disconnected from real-world needs, while data and feedback generated across application settings may fail to inform subsequent optimization and decision-making [10]. Second, medical innovation operates within stringent biosafety and ethical boundaries. A stakeholder model can identify potential compliance conflicts at an early stage and, through transparent collaborative mechanisms, provide legitimacy for the equitable social diffusion of emerging technologies [11].

2. Stakeholder Mapping

Within the theoretical framework described above, the first question is which stakeholders are involved in the translation of ByeGerm and what roles and relative positions these actors occupy within the project. Stakeholder theory provides a basic analytical perspective for identifying relevant actors, but a purely textual list cannot adequately represent hierarchical differences and relational structures among multiple stakeholders. This study therefore converts the stakeholder-identification results into a spatial stakeholder map to provide an integrated representation of the stakeholder composition and relative positioning throughout the ByeGerm translation process.

Traditional stakeholder maps, however, are often based primarily on qualitative judgment. Consequently, assessments of actor importance and hierarchical placement may be strongly influenced by researcher subjectivity, with common problems including poor indicator operationalization, insufficient justification for weighting, and the absence of objective boundaries between layers [22, 17]. To enhance the academic rigor and decision robustness of the stakeholder map, this study adopts a progressive analytical procedure.

First, existing stakeholder theories are reviewed to establish the theoretical basis for the project. A stakeholder-identification mechanism is then developed to narrow the initially unordered universe of social actors to project-relevant stakeholders. Stakeholder analysis is subsequently conducted to complete the topological mapping of actors within a concentric-circle space and to examine their ecological relationships.

2.1 Establishing the Three-Dimensional Theoretical Framework

Since the stakeholder concept was introduced, numerous theoretical models have been developed to identify and manage multiple actors, and the major approaches have accumulated substantial empirical foundations [12]. Freeman's seminal work first distinguished stakeholders from shareholders and established the broad, bidirectional ontological boundary of "affecting and being affected," which became a foundational premise for subsequent classification models [13]. In the 1990s, stakeholder theory increasingly evolved toward classification matrices. Savage et al. proposed the threat-cooperation matrix, which classifies stakeholders according to their potential threat to a project and their willingness to cooperate, thereby supporting defensive, monitoring, and collaborative strategies; this model has been widely applied in highly adversarial business contexts [14]. Frederick subsequently introduced a direct-indirect distinction from the perspective of organizational boundaries, separating groups embedded in business transaction chains from broader environmental groups that exert policy or public-opinion constraints. This distinction became an important approach in public-affairs analysis [15]. Evaluation dimensions were later further refined. Based on resource-dependence theory, Clarkson proposed the primary-secondary stakeholder model, which focuses on the degree to which organizational survival depends on particular groups; this approach has been widely applied in resource-constrained entrepreneurial ecosystems [16]. During the same period, Mitchell et al. introduced the stakeholder salience model, using the dynamic attributes of power, legitimacy, and urgency to assess stakeholder priority; the model has become influential in crisis and emergency management [17, 18]. Rowley moved beyond dyadic relationships by incorporating social-network analysis and quantifying complex interactions through network density and centrality, thereby laying a foundation for stakeholder network topology [19].

The implementation of portable diagnostic devices in primary-level settings is characterized by stringent compliance requirements, long translational cycles, and a strong orientation toward social accessibility. An appropriate model must therefore accommodate the distinctive three-layer structure of the healthcare sector: policy and regulation, commercial distribution, and clinical application. On this basis, the present study screened existing stakeholder models. Although the salience model is well suited to emergency contexts, the present project focuses on routine, system-level translation; its emphasis on short-term "urgency" is therefore less compatible with the project's needs. The Savage matrix is also grounded in an adversarial-game assumption, whereas healthcare and public-health ecosystems place greater emphasis on multi-actor value co-creation. The project therefore selected three theories with strong contextual fit and integrated their theoretical cores into a composite framework.

First, Freeman's broad foundational theory was selected. Its underlying principle is that no relevant party should be omitted. This logic ensures that marginalized or vulnerable groups that are easily overlooked by conventional commercial models, such as older adults in nursing homes and children in childcare settings, are considered on an equal basis when planning real-world implementation, thereby providing a strong ethical foundation for equitable access [20, 7].

Second, Frederick's direct-indirect dichotomy was adopted. Medical products are subject to stringent market-authorization procedures and price controls. This model clearly distinguishes between groups that participate directly across the research, development, and production process and have direct business interactions, and groups that provide indirect enabling or constraining effects through policy orientation

and public opinion. This distinction closely matches the practical need for differentiated engagement strategies across different stages of the project [15].

Third, Clarkson's primary-secondary dependence model was selected. This theory focuses on core partners whose participation is indispensable to the survival and development of a project. At an early stage characterized by severe resource-allocation trade-offs, it provides a theoretical basis for prioritizing end-user experience and the public-health value of disease prevention and control [16].

By integrating these three theoretical perspectives and considering the context-specific implementation requirements of portable diagnostic devices, this study develops a three-dimensional assessment framework comprising degree of association, depth of participation, and directness of influence. Degree of association measures how closely an actor is tied to the project's core activities; depth of participation captures the extent of involvement across the full value chain; and directness of influence distinguishes the pathways through which each actor affects the realization of project value.

2.2 Stakeholder Identification

In the complex ecosystem of medical-technology translation, accurately defining system boundaries and identifying stakeholders without systematic omission or bias are prerequisites for quantitative modeling [23]. Conventional qualitative identification methods are susceptible to the limits of researcher cognition and may therefore omit relevant actors or produce ambiguous boundaries [24]. To address this problem, the present study applies Boolean touchpoint filtering to derive a target stakeholder set from the unordered universe of social actors [25].

Let the set of all potential entities in society be represented by the unordered universe U. To determine mathematically whether an entity $x \in U$ has a substantive system-level intersection with the portable diagnostic technology platform, a Boolean touchpoint evaluation vector $b(x) = \left(b_{1(x)}, b_{2(x)}, b_{3(x)}, b_{4(x)}\right)^T \in \{0,1\}^4$ is constructed [26]. The four Boolean primitive conditions are defined as follows:

$b_1(x) \in \{0,1\}$ Biological-contact domain: whether entity x is a potential host of project-related pathogens, a susceptible population, a high-risk exposed population, or a setting in which concentrated prevention and control measures are implemented.

$b_2(x) \in \{0,1\}$ Business-process domain: whether entity x participates in the research and development, processing, transportation, sales, clinical validation, diagnostic delivery, or health-insurance settlement associated with the diagnostic device.

$b_3(x) \in \{0,1\}$ Administrative-constraint domain: whether entity x holds authority over market-access approval, quality supervision, public-health emergency allocation, or policy formulation related to the project product.

$b_4(x) \in \{0,1\}$ Social-communication domain: whether entity x is involved in ethical discussion, public understanding, or trust formation related to the project.

Define the Boolean indicator function for project interaction touchpoints as $I_{touch}(x) \in \{0,1\}$. Its Boolean output equals 1 if and only if entity x satisfies at least one of the touchpoint conditions above.

Through Boolean algebra, entities with zero association under $I_{touch}(x) = 0$ are excluded, yielding the effective candidate set $S \subset U$:.

$$S = \{x \in U | I_{\text{touch}}(x) = 1\}$$

The preceding screening step resolves the system-boundary problem, but entities within the candidate set S remain heterogeneous. To achieve mutually exclusive and collectively exhaustive categorization, a four-dimensional functional-role attribute vector $\mathbf{F}(x) = \left(f_{biological}, f_{commercial}, f_{regulatory}, f_{societal}\right)^T \in \{0,1\}^4$ is established [27].

Based on the mutually exclusive and complementary mapping of functional attributes $\mathcal{M}: S \rightarrow \{C_1, C_2, ..., C_{13}\}$, the candidate set S is partitioned into non-overlapping subsets C1, C2, ..., C13 such that their union equals S and Ci ∩ Cj = ∅ for all i ≠ j.

Using this two-stage mathematical framework, the study reduces the complex social network to 14 target stakeholder categories with clearly defined boundaries and functions.

1. Patients

Patients are key end users of respiratory infectious-disease testing services. Their requirements regarding diagnostic accuracy, turnaround time, ease of operation, and comprehensibility of results directly shape the functional positioning and product design of convenient rapid-testing technologies. The experiences and feedback of patients and caregivers during actual use provide direct evidence for optimizing sampling procedures, operating methods, result presentation, and device interaction, and therefore represent an important source of continuous technological iteration. Subject to privacy-protection and data-governance requirements, standardized test results can also provide supporting information for clinical judgment, infection prevention and control, and public-health surveillance. Patients are therefore not only recipients of testing services but also key stakeholders in determining the usability, clinical value, and practical utility of convenient rapid-testing technologies [28-31].

2. Susceptible Populations

Susceptible populations are a priority group for early screening and risk prevention of respiratory infectious diseases. Their greater need for early infection identification and timely testing drives the extension of convenient, rapid, low-threshold testing services from traditional medical institutions to homes, communities, and other everyday settings. Vulnerable groups such as children and older adults have particularly stringent requirements regarding sampling experience, ease of operation, readability of results, and safety. In response to these real-world needs, the project iteratively optimizes the testing workflow and human-machine interaction design so that the device better matches the operational capabilities and use environments of priority populations, facilitates rapid identification of infection risk, and supports early intervention and primary-level prevention and control of respiratory infectious diseases [32, 33].

3. Close Contacts

Close contacts are a priority population in the prevention and control of respiratory infectious-disease transmission chains, and their testing needs are typically highly time-sensitive. During cluster outbreaks or specific infection events, close contacts may also generate temporary demand for large-scale concentrated testing, creating higher requirements for testing speed, operational efficiency, and batch-processing capacity. The practical needs of this group therefore require the project to maintain a strong focus on workflow efficiency and on-site applicability so that the technology can respond effectively to real-world use conditions [34, 35].

4. Nursing Homes and Kindergartens

Nursing homes and kindergartens are settings in which older adults and children live or participate in activities collectively, and they are also important sites for preventing and controlling clustered transmission of respiratory infectious diseases. Such settings place particularly high demands on testing speed, ease of operation, sampling experience, and on-site convenience, prompting the project to optimize devices and workflows from the perspective of actual use. Frontline feedback further helps identify age-specific barriers in sampling, operation, and result interpretation and supports the continuous improvement of devices, operating instructions, and use procedures. At the same time, provided that data-governance and public-health requirements are met, testing information generated in these settings can support infection surveillance and prevention and control [36, 37].

5. Public Health Institutions

Public health institutions perform critical functions in the surveillance, early warning, and control of respiratory infectious diseases. Their requirements for testing quality, standardized results, data governance, and information interoperability directly influence the data-management and application models of convenient rapid-testing technologies. Field applications and population screening conducted with the participation of public health institutions can provide evidence regarding device feasibility, suitability, and implementation conditions in real public-health environments. In addition, requirements relating to data quality, result reporting, and information security can promote more standardized integration between convenient testing technologies and existing public-health surveillance systems [38, 39].

6. Healthcare Institutions

Healthcare institutions are important professional stakeholders in the diagnosis and treatment system for respiratory infections. They play a key role in professional guidance for rapid testing outside the hospital, clinical evaluation, and linkage to subsequent diagnosis and treatment. Their requirements for testing accuracy, result interpretation, clinical applicability, and standardized use provide an important basis for optimizing convenient rapid-testing technologies. Professional feedback from healthcare personnel helps the project define appropriate out-of-hospital use scenarios and application boundaries and continuously improve device and workflow design so that the technology can better support risk identification outside the hospital and subsequent medical decision-making [40, 41].

7. Researchers in Relevant Fields

Researchers in relevant fields include specialists in respiratory infection, pathogen detection, molecular diagnostics, laboratory medicine, artificial intelligence, epidemiology, and public health. They

are important stakeholders linking technology development, scientific validation, and real-world application. Researchers can provide professional guidance throughout the process, from technical principles and system construction to performance validation and practical application, helping the project assess the appropriateness of technical pathways, improve experimental and evaluation protocols, and identify key issues across different application settings. As the project increasingly enters real-use environments, researchers can also use their expertise to evaluate device stability, reliability of testing results, operational procedures, and conditions of use, thereby informing technological optimization and contextual adaptation. Multidisciplinary research participation enables the project to advance technological development while maintaining scientific rigor, reliability, and responsiveness to practical needs, and provides professional support for continued device iteration and application expansion [42, 43].

8. Manufacturing Partners

Manufacturing partners are critical actors in translating portable rapid-testing devices from research prototypes into commercial products. Their manufacturing processes, quality-control capabilities, batch consistency, and supply-chain reliability directly determine product quality, production cost, and the potential for large-scale manufacturing. Moving a technology from the laboratory to real-world application requires attention not only to diagnostic performance but also to engineering feasibility, quality systems, resource allocation, and implementation cost. Practical feedback from manufacturing partners on structural design, material selection, assembly processes, and production-line workflows can help identify the gap between laboratory prototypes and industrial-scale production and support iterative development toward a form that combines stability, compliance, and manufacturability. Collaboration with manufacturing partners therefore determines not only whether stable mass production can be achieved, but also the subsequent deployment of the device in primary-level and public-health settings [44, 45].

9. Distributors

Distributors are responsible for product circulation, channel expansion, and selected after-sales services and are therefore important market actors in bringing portable rapid-testing technologies into diverse real-world application settings. Their contributions to product supply, user training, after-sales support, and market-information feedback directly affect product accessibility and practical implementation efficiency [46].

10. Health Insurance Payers

Health insurance payers influence the economic accessibility of medical services and related diagnostic products through reimbursement policies, access mechanisms, and settlement arrangements. For convenient rapid-testing technologies for respiratory infectious diseases, payment methods and cost-sharing mechanisms may affect actual uptake by healthcare institutions and patients. Stakeholder research indicates that well-designed reimbursement and payment policies can reduce financial barriers for target populations and facilitate implementation of diagnostic technologies. Accordingly, payer requirements may further shape product cost control, pricing strategies, application settings, and adoption pathways within healthcare institutions [47, 48].

11. The General Public

The general public constitutes an important social foundation for the adoption of convenient rapid-testing technologies. Public understanding of respiratory infectious diseases, trust in test results, and concerns regarding privacy and data security can influence social acceptance of these technologies. Public expectations concerning testing convenience, reliability of results, cost of use, and information transparency can in turn drive continuous improvement in user experience, result interpretation, privacy protection, and communication [49, 50].

12. Educational Institutions

Educational institutions include schools, community education platforms, and science-communication organizations and play an important role in improving public health literacy and understanding of diagnostic testing. Health education and science communication can help the public understand the transmission characteristics of respiratory infectious diseases, the purpose of testing, and the basic meaning of test results, thereby reducing comprehension and use barriers arising from insufficient knowledge. Educational institutions can also communicate cognitive and operational feedback from groups such as children and adolescents to the project team, informing the development of population-appropriate science-communication materials, operating guidance, and information-presentation strategies [51, 52].

13. Regulatory Authorities

Regulatory authorities responsible for drug and medical-device administration, health, and related functions are important stakeholders in the development and real-world implementation of portable rapid-testing technologies. Through systems governing medical-device registration, clinical evaluation, product quality, manufacturing and distribution, and use oversight, regulators establish standards for product safety, effectiveness, and compliance. These institutional requirements define the access threshold for real-world application and drive continuous improvement across the product life cycle, including performance validation, safety assurance, and quality management [53, 54].

14. Policymakers

Policymakers provide high-level direction for the development of the public-health sector and infectious-disease prevention systems. Medical and epidemic-prevention standards, public-health and social-welfare policies, and science-and-innovation support measures establish the project's compliance framework and boundaries for real-world implementation. Relevant industry guidelines and primary-level prevention requirements define standards for the research and development, mass production, and market access of portable diagnostic devices, thereby directing project iteration toward pressing public-health needs. Policy support for home testing, routine primary-level screening, and public-health emergency preparedness also facilitates deployment in communities, schools, nursing facilities, and other settings and broadens potential application and dissemination pathways. In addition, high-level epidemic-prevention planning and public-health policies oriented toward social benefit require the project to balance practicality, accessibility, and safety, thereby improving alignment with routine public-health prevention systems and enhancing both social value and implementation feasibility [55, 56].

Code	C_i	$\mathbf{b}(x)(b_1, b_2, b_3, b_4)$	$\mathbf{F}(x)$
C_1	Patients	(1, 0, 0, 1)	(1, 0, 0, 0)
C_2	Susceptible populations	(1, 0, 0, 1)	(1, 0, 0, 1)
C_3	Close contacts	(1, 0, 0, 1)	(1, 0, 0, 1)
C_4	Nursing homes and kindergartens	(1, 1, 0, 1)	(1, 0, 0, 1)
C_5	Public health institutions	(1, 0, 1, 1)	(0, 0, 1, 1)
C_6	Healthcare institutions	(1, 1, 1, 0)	(1, 0, 0, 0)
C_7	Manufacturing partners	(0, 1, 1, 0)	(0, 1, 0, 0)
C_8	Distributors	(0, 1, 0, 0)	(0, 1, 0, 0)
C_9	Health insurance payers	(0, 1, 1, 0)	(0, 1, 0, 0)
C_{10}	Regulatory authorities	(0, 0, 1, 0)	(0, 0, 1, 0)
C_{11}	Policymakers	(0, 0, 1, 1)	(0, 0, 1, 1)
C_{12}	Researchers in relevant fields	(1, 0, 1, 0)	(1, 0, 1, 0)
C_{13}	Educational institutions	(1, 0, 0, 0)	(1, 0, 0, 0)
C_{14}	General public	(0, 0, 0, 1)	(0, 0, 0, 1)

2.3 Construction of the Stakeholder Map

2.3.1 Stakeholder Evaluation Criteria

This study uses degree of association, directness of influence, and continuity dependence as the three primary dimensions for quantitative stakeholder evaluation. Based on the practice of the ByeGerm project, these primary dimensions are further decomposed into secondary evaluation factors and tertiary observable indicators, thereby operationalizing theoretical concepts as measurable variables.

Primary dimension	Secondary evaluation factor	Tertiary observable indicator	Measurement meaning
Degree of association (A)	Task association (A1)	Core-task association (A11)	Degree of linkage between the stakeholder and core activities in respiratory-infection diagnosis and treatment, public-health prevention and control, and health services
		Technology-need association (A12)	Degree of match between the diagnostic need addressed by ByeGerm and the stakeholder's actual health, service, or operational needs
	Risk association (A2)	Health-risk exposure (A21)	Degree to which the stakeholder is exposed to health risks during respiratory-infection transmission, diagnosis, treatment, and prevention and control
		Impact on stakeholder objectives (A22)	Degree to which the effectiveness or absence of the diagnostic technology affects the stakeholder's health, service, or organizational objectives
Directness of influence (I)	Directness of behavioral intervention (I1)	Directness of testing and control action (I11)	Degree to which stakeholder behavior directly affects testing implementation, diagnostic or treatment judgment, or prevention-and-control measures
		Influence on application decisions (I12)	Degree to which the stakeholder influences decisions on technology access, procurement, allocation, use, or application
	Directness of end-use impact (I2)	Directness of end-user service (I21)	Degree to which the stakeholder directly provides products, services, technical support, or related resources to testing populations or end-use settings
		Directness of diffusion impact (I22)	Degree to which the stakeholder directly promotes technology adoption, contextual application, and further diffusion
Continuity dependence (S)	Functional dependence (S1)	Health-protection dependence (S11)	Degree to which the stakeholder's health protection depends on testing, diagnosis and treatment, and public-health prevention-and-control services
		Service-function dependence (S12)	Degree to which the stakeholder depends on the testing system to perform its responsibilities or maintain relevant service functions
	Sustained coupling (S2)	Sustained life-cycle participation (S21)	Degree of sustained stakeholder participation across research and development, validation, production, application, regulation, and public-health response
		Collaborative dependence (S22)	Degree of sustained information, resource, service, or decision coordination between the stakeholder, the project, and other stakeholders

To ensure that different stakeholders can be compared on a common scale, the study evaluates the 12 tertiary observable indicators using a five-point Likert scale.

Tertiary observable indicator	1 point	2 points	3 points	4 points	5 points
Core-task association	Essentially no direct connection	Weak connection	Clear connection	Closely related to multiple core stages	Directly involved in core activities
Technology-need association (A12)	Essentially no relevant need	Weakly related need	Clear need	Technology clearly addresses the need	Technology need is direct and central
Health-risk exposure	Essentially unaffected by relevant risks	Low risk exposure	Moderate risk exposure	High risk exposure	Direct and sustained exposure to a high-risk environment
Impact on objectives	Essentially unaffected	Minor impact	Clear impact	Substantial impact	Direct and major impact on core objectives
Directness of testing and control action	Essentially no direct effect	Weak effect	Participates in selected relevant stages	Directly influences key stages	Directly implements or determines relevant actions
Influence on application decisions	Essentially no influence	Indirect influence only	Participates in some decisions	Clear influence on important decisions	Directly determines or strongly controls key decisions
Directness of end-user service	No direct contact with end users	Provides indirect support	Participates in some end-user services	Directly provides substantial services or resources	Directly provides core services to end users
Diffusion impact	Essentially no diffusion effect	Weak diffusion effect	Participates in promotion	Clearly promotes application and diffusion	Directly drives technology adoption and diffusion
Health-protection dependence	Essentially no dependence	Low dependence	Stable dependence	Health protection depends substantially on the system	Highly important to health protection
Service-function dependence	Essentially no dependence	Low dependence	Clear dependence	Operations depend substantially on the system	Core service functions are highly dependent
Sustained life-cycle participation	Occasional participation in a single stage	Limited participation	Sustained participation in selected stages	Involved across multiple key stages	Sustained participation across multiple key stages
Collaborative dependence	Essentially no sustained coordination	Limited coordination	Stable coordination	Sustained multidimensional coordination	Highly sustained dependence involving resources, information, or services

2.3.2 Delphi Expert Scoring and Formation of Raw Tertiary-Indicator Data

Using a triangulation approach, the team invited 12 project-team members, 6 experts in public health, 6 experts in medicine, and 12 target users (including hospital patients, representatives of nursing homes and kindergartens, and members of the general public) to participate in scoring through the Delphi expert-consultation method [57, 58].

The coefficient of variation $CV_{ij} = \frac{s_{ij}}{\bar{x}_{ij}}$ was used to measure the dispersion of expert opinions, with $CV_{ij} \leq 0.25$ adopted as the convergence criterion for each individual score. Kendall's coefficient of concordance was also used to assess the consistency of the overall expert-ranking structure, together with significance testing. In the first round, the mean CV across the 12 tertiary indicators and 14 stakeholder categories was 0.149, with a maximum of 0.333, while Kendall's W was 0.815 ($p < 0.001$). After the statistical results were fed back to the experts, a second round of evaluation was conducted. The mean CV decreased to 0.105 and the maximum CV decreased to 0.236, satisfying the convergence requirement of $CV \leq 0.25$; Kendall's W further increased to 0.899 ($p < 0.001$). The second-round results were therefore adopted as the final expert-evaluation data [58].

The final score x_{ij} of stakeholder category i on tertiary indicator j was calculated as the arithmetic mean of expert ratings: $x_{ij} = \frac{1}{N}\sum_{r=1}^{N} x_{ijr}$

where N = 10 is the number of experts participating in the evaluation, and x_{ijr} denotes the score assigned by expert r to stakeholder category i on tertiary indicator j. This yields the raw tertiary-indicator evaluation matrix 14×12: $\mathbf{X}^{(3)} = \left(x_{ij}\right)_{14\times12}$

The table below presents the aggregated matrix from the second round of expert evaluation.

Stakeholder	A11	A12	A21	A22	I11	I12	I21	I22	S11	S12	S21	S22
Patients	5	5	5	5	5	4	5	4	4	4	4	4
Susceptible populations	5	5	5	5	4	4	4	4	4	4	4	4
Close contacts	5	5	5	5	5	5	4	4	4	4	4	4
Nursing homes and kindergartens	5	5	5	5	5	5	5	4	4	4	4	4
Healthcare institutions	5	4	4	5	5	5	5	5	5	4	5	5
Public health institutions	5	4	5	5	5	5	5	5	5	5	5	5
Researchers in relevant fields	4	4	3	4	4	4	4	4	4	4	4	4
Distributors	3	3	2	3	3	3	3	5	4	4	3	3
Manufacturing partners	3	3	2	3	3	3	4	5	5	4	3	4
Health insurance payers	3	3	2	3	3	3	3	4	4	5	5	4
Policymakers	2	2	1	2	3	3	2	2	2	3	5	5
Regulatory authorities	2	2	1	2	3	3	2	2	2	3	5	5
General public	3	3	3	3	3	3	2	2	2	2	4	4
Educational institutions	3	3	3	4	3	3	2	2	2	2	4	4

2.3.3 Aggregation from Tertiary to Secondary Indicators

The 12 tertiary indicators are grouped under six secondary evaluation factors. Because the two tertiary indicators within each secondary factor represent parallel measurements, they are aggregated using the arithmetic mean.

This procedure yields the six-factor secondary-indicator evaluation matrix $\mathbf{X}_{14\times6}$ for the 14 stakeholder categories.

Stakeholder	A1	A2	I1	I2	S1	S2
Patients	5	5	4.5	4.5	4	4
Susceptible populations	5	5	4	4	4	4
Close contacts	5	5	5	4	4	4
Nursing homes and kindergartens	5	5	5	4.5	4	4
Healthcare institutions	4.5	4.5	5	5	4.5	5
Public health institutions	4.5	5	5	5	5	5
Researchers in relevant fields	4	3.5	4	4	4	4
Distributors	3	2.5	3	4	4	3
Manufacturing partners	3	2.5	3	4.5	4.5	3.5
Health insurance payers	3	2.5	3	3.5	4.5	4.5
Policymakers	2	1.5	3	2	2.5	5
Regulatory authorities	2	1.5	3.5	2.5	2	5
General public	3	3	3	2	2	4
Educational institutions	3	3.5	3	2	2	4

2.3.4 Min-Max Normalization

Because all six secondary indicators are positive benefit-type indicators, Min-Max range normalization is applied column-wise to obtain dimensionless standardized values.

$$z_{ik} = \frac{x_{ik} - \min\limits_{i}(x_{ik})}{\max\limits_{i}(x_{ik}) - \min\limits_{i}(x_{ik})}$$

where x_{ik} is the raw score of stakeholder category i on secondary indicator k, and $\min\limits_{i}(x_{ik})$ and $\max\limits_{i}(x_{ik})$ are respectively the minimum and maximum values of that indicator across the 14 stakeholder categories. The standardized matrix $\mathbf{Z}_{14\times6}$ is shown in the table below and satisfies $\min(z_k) = 0, \max(z_k) = 1$ for each column.

Stakeholder	A1	A2	I1	I2	S1	S2
Patients	1.000	1.000	0.750	0.833	0.667	0.500
Susceptible populations	1.000	1.000	0.500	0.667	0.667	0.500
Close contacts	1.000	1.000	1.000	0.667	0.667	0.500

Stakeholder	A1	A2	I1	I2	S1	S2
Nursing homes and kindergartens	1.000	1.000	1.000	0.833	0.667	0.500
Healthcare institutions	0.833	0.857	1.000	1.000	0.833	1.000
Public health institutions	0.833	1.000	1.000	1.000	1.000	1.000
Researchers in relevant fields	0.667	0.571	0.500	0.667	0.667	0.500
Distributors	0.333	0.286	0.000	0.667	0.667	0.000
Manufacturing partners	0.333	0.286	0.000	0.833	0.833	0.250
Health insurance payers	0.333	0.286	0.000	0.500	0.833	0.750
Policymakers	0.000	0.000	0.000	0.000	0.167	0.000
Regulatory authorities	0.000	0.000	0.250	0.167	0.000	1.000
General public	0.333	0.429	0.000	0.000	0.000	0.500
Educational institutions	0.333	0.571	0.000	0.000	0.000	0.500

2.3.5 AHP Subjective Weighting

Based on expert judgments of the relative importance of the six secondary indicators, the pairwise-comparison judgment matrix $\mathbf{A} = (a_{kl})_{6\times6}$ is constructed [59].

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1/2 & 1 & 1/2 & 1 \\ 1 & 1 & 1/2 & 1 & 1/2 & 1 \\ 2 & 2 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1/2 & 1 & 1/2 & 1 \\ 2 & 2 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1/2 & 1 & 1/2 & 1 \end{pmatrix}$$

The geometric-mean method is used to calculate the subjective weight vector $\boldsymbol{w}^{AHP}$: $u_k = \left(\prod_{l=1}^{6} a_{kl}\right)^{1/6}, \quad w_k^{AHP} = \frac{u_k}{\sum_{k=1}^{6} u_k}$

The judgment matrix is then subjected to a consistency test: $\lambda_{\max} = 6.000$

$$CI = \frac{\lambda_{\max} - 6}{6 - 1} = 0.000$$

$$CR = \frac{CI}{RI} = \frac{0.000}{1.24} = 0.000 < 0.10$$

where RI = 1.24 is the average random consistency index for a sixth-order judgment matrix. The results indicate that the judgment matrix is perfectly consistent.

Indicator	A1	A2	I1	I2	S1	S2
$\boldsymbol{w}^{AHP}$	0.125	0.125	0.250	0.125	0.250	0.125

2.3.6 EWM Objective Weighting

Based on the standardized matrix $\mathbf{Z}_{14\times6}$, the entropy weight method (EWM) is used to calculate the objective weight of each indicator.

Following [60], the information proportion of stakeholder category i under indicator k is calculated as follows: $p_{ik} = \frac{z_{ik}}{\sum_{i=1}^{14} z_{ik}}$

The information entropy e_k of each indicator is then calculated: $e_k = -\frac{1}{\ln 14}\sum_{i=1}^{14} p_{ik} \ln p_{ik}$

When $p_{ik} = 0$, the corresponding entropy term is defined as 0 by continuity.

The information utility value d_k and objective weight w_k^{EWM} are then calculated: $d_k = 1 - e_k$, $w_k^{EWM} = \frac{d_k}{\sum_{k=1}^{6} d_k}$

Indicator	Entropy	Information utility	EWM objective weight
A1	0.9019	0.0981	0.1369
A2	0.9023	0.0977	0.1364
I1	0.7582	0.2418	0.3373
I2	0.8855	0.1145	0.1597
S1	0.8912	0.1088	0.1518
S2	0.9441	0.0559	0.0779

2.3.7 AHP-EWM Geometric-Mean Combination Weighting

To integrate expert judgment with the intrinsic dispersion information contained in the sample data, the geometric-mean method is used to combine the AHP subjective weights and EWM objective weights [61].

$$\tilde{w}_k = \sqrt{w_k^{AHP} \cdot w_k^{EWM}}$$

After normalization, the final combined weight vector $\boldsymbol{\omega} = (\omega_1, \omega_2, ..., \omega_6)$: is obtained.

$$\omega_k = \frac{\tilde{w}_k}{\sum_{k=1}^{6} \tilde{w}_k}$$

Indicator	$AHPweightw^{AH}$	$EWMweightw^{EWI}$	Geometric mean	Combined weight
A1	0.125	0.1369	0.1326	0.1326
A2	0.125	0.1364	0.1323	0.1323
I1	0.250	0.3373	0.2944	0.2944

Indicator	$AHPweightw^{AH}$	$EWMweightw^{EW}$	Geometric mean	Combined weight
I2	0.125	0.1597	0.1432	0.1432
S1	0.250	0.1518	0.1974	0.1974
S2	0.125	0.0779	0.1000	0.1000
Total	1.000	1.000	1.0000	1.0000

2.3.8 MAUT Composite Utility Calculation and Radial Mapping

Based on the combined weight vector $\boldsymbol{\omega}$ and standardized matrix $\mathbf{Z}_{14\times6}$, multi-attribute utility theory (MAUT) is used to calculate the composite score S_i for each stakeholder category [62]:

$$S_i = \sum_{k=1}^{6} \omega_k \, z_{ik}$$

Clearly, $0 \leq S_i \leq 1$. The closer S_i is to 1, the greater the overall salience of the corresponding stakeholder within the A-I-S evaluation framework.

To convert the one-dimensional composite utility score into a visual radial coordinate for the concentric-circle representation, the following radial mapping function is defined:

$$\rho_i = 1 - S_i$$

ρ_i represents the relative radial coordinate of a stakeholder in the concentric-circle layout. A smaller ρ_i indicates a position closer to the center, whereas a larger ρ_i indicates a position farther from the center.

Stakeholder	Composite score	Mapped radial coordinate
Patients	0.7867	0.21333
Susceptible populations	0.6893	0.3107
Close contacts	0.8365	0.1635
Nursing homes and kindergartens	0.8603	0.1397
Healthcare institutions	0.9259	0.0741
Public health institutions	0.9778	0.0222
Researchers in relevant fields	0.5883	0.4117
Distributors	0.3092	0.6908
Manufacturing partners	0.3907	0.6093
Health insurance payers	0.3930	0.6070
Policymakers	0.1330	0.8670
Regulatory authorities	0.1975	0.8025
General public	0.1510	0.8490
Educational institutions	0.1697	0.8303

2.3.9 Polar Mapping and One-Dimensional K-Means Layer Classification

Considering the theoretically defined three-layer stakeholder governance structure of “core-direct-indirect” and the clustering-evaluation results obtained under different numbers of clusters, this study

applies one-dimensional K-means clustering with K = 3 to the 14 radial-coordinate values $\{\rho_1, \rho_2, \ldots, \rho_{14}\}$ [63].

The mathematical objective is to minimize the within-cluster sum of squared errors (SSE): $\min \sum_{j=1}^{3} \sum_{\rho_i \in C_j} (\rho_i - \mu_j)^2$

After iterative convergence, the three cluster centroids, in ascending order, are: $\mu_1 = 0.1539, \quad \mu_2 = 0.5797, \quad \mu_3 = 0.8372$

The boundary between each pair of adjacent clusters is determined by the midpoint between the radial coordinates of the adjacent boundary observations: $\tau_1 = \frac{\max(\rho_{i \in C_1}) + \min(\rho_{i \in C_2})}{2} = \frac{0.3107 + 0.4117}{2} = 0.3612$

$$\tau_2 = \frac{\max(\rho_{i \in C_2}) + \min(\rho_{i \in C_3})}{2} = \frac{0.6908 + 0.8025}{2} = 0.7467$$

According to the calculated threshold boundaries, the classification rules are as follows:

Core layer: $\rho_i < 0.3612$

Direct layer: $0.3612 \leq \rho_i < 0.7467$

Indirect layer: $\rho_i \geq 0.7467$

These classification results form the concentric-circle stakeholder structure.

2.3.10 Concentric-Circle Stakeholder Structure and Classification Results

Based on the quantitative calculations and objective clustering boundaries described above, stakeholders are classified as follows.

The core-stakeholder layer comprises public health institutions (0.0222), healthcare institutions (0.0741), nursing homes and kindergartens (0.1397), close contacts (0.1635), patients (0.2133), and susceptible populations (0.3107). The radial coordinates of stakeholders in this layer satisfy $\rho_i < 0.3612$, with a cluster centroid of $\mu_1 = 0.1539$. These stakeholders have the most direct operational relationships with, or health-related dependence on, pathogen screening, diagnostic services, and epidemic prevention and control. They are the immediate recipients of products and services and the frontline actors responsible for implementation.

The direct-stakeholder layer comprises researchers in relevant fields (0.4117), health insurance payers (0.6070), manufacturing partners (0.6093), and distributors (0.6908). The radial coordinates of this layer satisfy $0.3612 \leq \rho_i < 0.7467$, with a cluster centroid of $\mu_2 = 0.5797$. These stakeholders participate mainly through mechanisms such as technology development, manufacturing, supply-chain distribution, and financial payment, providing direct technical and resource support to the core layer.

The indirect-stakeholder layer comprises regulatory authorities (0.8025), educational institutions (0.8303), the general public (0.8490), and policymakers (0.8670). The radial coordinates of this layer satisfy $\rho_i \geq 0.7467$, with a cluster centroid of $\mu_3 = 0.8372$. These stakeholders do not directly participate in specific testing or production processes; rather, they exert macro-level influence on the overall system through external environmental factors such as regulation, industry oversight, public opinion, and health education.

2.4 Ecological-Niche Analysis Based on the Stakeholder Map

(1) Integration of Multiple Roles among Core Stakeholders

Within the application ecosystem for respiratory-disease detection technologies, patients, susceptible populations, close contacts, healthcare institutions, nursing homes and kindergartens, and public health institutions form an interconnected, data-driven collaborative network. Centered on testing data, these six stakeholder categories constitute an integrated system of "demand initiation - technological response - data integration - prevention-and-control feedback" [71].

As frontline users, patients, susceptible populations, and close contacts obtain individual health information through testing. Relevant data can be transmitted to healthcare institutions to support more accurate diagnosis and treatment and can also be shared, where appropriate, with nursing homes and kindergartens to facilitate proactive risk screening in priority settings.

After receiving testing data, healthcare institutions serve simultaneously as clinical and information hubs. They provide individualized medical services while transmitting key epidemiological information to public health institutions in a timely manner, thereby supporting regional risk assessment and decision-making.

Public health institutions integrate multisource data from healthcare institutions, nursing homes, and kindergartens to assess outbreaks, trace transmission chains, and formulate prevention-and-control strategies. Early-warning information can then be communicated back to frontline settings to guide dynamic optimization of testing and prevention procedures.

After rapid-testing technologies are introduced into nursing homes and kindergartens, these institutions can provide older adults and children with convenient testing close to where they live or study and can use testing data to support routine health monitoring. On-site test results can be communicated to healthcare personnel to assist preliminary clinical assessment and health management, while healthcare personnel can provide guidance on standardized operation, workflow optimization, and emergency response to reduce the risk of clustered transmission. Relevant testing data and abnormal findings can also be reported, in accordance with applicable requirements, to local public health institutions, providing frontline evidence for surveillance, early warning, and targeted prevention and control. Screening frequency and protective measures can then be adjusted in response to public-health alerts, creating a closed-loop management system that links institutional self-screening, professional medical guidance, and public-health oversight [64].

Through mutual dependence and data exchange, these six stakeholder categories extend the value of testing technologies from individual health management to public-health emergency prevention and control, enabling a transition from isolated testing to system-level protection.

(2) Closed-Loop Commercial Collaboration among Direct Stakeholders

Within the implementation ecosystem, researchers in relevant fields, health insurance payers, manufacturing partners, and distributors form a closely connected industry-academia-research-commerce collaborative system [65] that helps move synthetic-biology designs from the laboratory into real-world use.

Pricing and reimbursement policies established by health insurance payers influence the entire value chain. Manufacturing partners use qualifications such as medical-device registration certificates to support access to reimbursable markets and employ lean manufacturing to address cost pressures created by reimbursement pricing. Distributors conduct market promotion within the reimbursement framework and provide frontline feedback that can improve the reimbursement experience. Health insurance payers, in turn, can use distribution data to assess the value and impact of project outcomes [66, 72].

As contributors to technological iteration and evidence generation, researchers in relevant fields continuously provide advanced theoretical input, performance-optimization strategies, and real-world research data. Their work supplies a scientific basis for reimbursement pricing, product improvement, and market adaptation and helps bridge the knowledge gap between laboratory innovation and clinical and policy implementation.

Manufacturing partners function as engines of translation. Together with project researchers, they convert designs into products suitable for batch production, undertake manufacturing implementation, provide technical transfer and training support, and establish production-and-distribution alliances with distributors. Manufacturing capacity and delivery capability support market expansion, while real-world feedback from the market informs further product iteration. In this relationship, manufacturing partners and distributors share both risks and benefits.

The four stakeholder groups jointly respond to requirements associated with centralized procurement. Researchers continue to improve scientific validity and clinical value; manufacturing partners refine production processes and improve cost-effectiveness; and distributors ensure broad product supply and implementation-service coverage. Together, these efforts support the inclusion of the device within reimbursement systems at an affordable price and create a complete pathway from innovative research and industrial translation to socially accessible application of synthetic-biology technologies.

(3) Organic Linkage between Direct and Core Stakeholders

Health insurance payers, manufacturing partners, distributors, and researchers in relevant fields work closely with core stakeholders to establish an integrated network spanning technology translation, contextual application, and value delivery, thereby facilitating the entry of synthetic-biology innovation into clinical practice and supporting population health.

Grounded in technical principles, evidence-based findings, and developments at the public-health frontier, researchers in relevant fields provide theoretical support and scientific guidance throughout policy

alignment, product development, contextual implementation, and market dissemination. They also provide standardized evidence and optimization strategies for collaboration among multiple stakeholders, helping bridge basic research, industrial translation, and public application.

Health insurance payers can include testing devices suitable for eldercare and early-childhood settings within reimbursement catalogues and establish appropriate reimbursement proportions, thereby reducing the cost burden on nursing homes and kindergartens [67, 78] and improving the affordability of testing for older adults and children. Data on testing frequency, population coverage, and reimbursement demand can in turn inform policy optimization for these special settings. For patients, susceptible populations, and close contacts, reimbursement directly lowers financial barriers, while their needs and feedback provide evidence for payer policy adjustment. Reimbursement-compliance requirements can encourage healthcare institutions to standardize testing practices, and insurance settlement provides financial support to participating institutions; clinical data and service needs from healthcare institutions can then help payers refine reimbursement policy and compliance standards. Broader reimbursement coverage can expand testing uptake and provide a larger data foundation for public health surveillance and epidemic control, while public-health requirements and testing standards can guide the inclusion of priority tests within reimbursement arrangements. Researchers in relevant fields can further support policy evaluation, access assessment, and optimization of reimbursement standards through real-world data and evidence-based research, improving the scientific basis and appropriateness of insurance alignment.

The specific testing needs, physical constraints, and application feedback of nursing homes and kindergartens can guide manufacturing partners in optimizing product design and production processes, while manufacturers provide these settings with accurate, stable, and easy-to-operate devices. Manufacturing partners also support reliable testing services for patients, susceptible populations, and close contacts, whose user feedback can inform product iteration. For healthcare institutions, manufacturing partners respond to context-specific clinical needs by translating requirements for portability, efficiency, and ease of operation into technical improvements; clinical data generated by healthcare personnel can also inform product development. Public-health prevention needs provide manufacturers with direction for research and development, while manufacturers supply public health institutions with high-throughput and efficient testing equipment to support surveillance and risk assessment. Researchers in relevant fields contribute knowledge at the frontier of synthetic biology and insight into context-specific problems, helping manufacturers refine processes, optimize performance, and standardize product design so that the technology can be adapted to diverse implementation settings.

Through their channel networks, distributors deliver testing devices to nursing homes and kindergartens and provide after-sales service, technical support, and operational training. Feedback from these settings helps distributors optimize channel configuration. Distributors can also reduce access barriers for patients, susceptible populations, and close contacts, while communicating collected demand information to manufacturing partners and health insurance payers. In healthcare settings, distributors collaborate with medical personnel on training and promotion to accelerate clinical adoption; recommendations from healthcare professionals can support market expansion, while after-sales support addresses institutional use problems. Distributors can also respond rapidly to procurement and allocation needs from public health institutions, ensuring equipment supply for prevention-and-control activities,

while such procurement needs create market opportunities. Researchers in relevant fields provide professional content for marketing, science communication, training, and contextual adaptation, helping standardize communication, strengthen implementation professionalism, and support efficient and equitable diffusion of the technology.

Through this multi-stakeholder collaborative network - combining scientific input from researchers, policy support from health insurance, industrial translation by manufacturers, channel implementation by distributors, and contextual feedback from application stakeholders - technological advances can continuously generate public-health value through interaction with the healthcare system [73].

(4) Interrelationships among Indirect Stakeholders

Indirect stakeholders do not participate directly in core applications or routine data flows. Instead, they form a collaborative network through policy guidance, regulatory constraints, public understanding, and contextual adaptation.

Policymakers and regulatory authorities form the two high-level pillars of project compliance. Based on public-health security needs, policymakers introduce incentives such as technology-promotion subsidies and health-insurance inclusion, thereby defining regulatory direction and standards. Regulatory authorities implement technology-access review and supervise application processes, feeding implementation problems back to policymakers and thereby supporting iterative policy improvement. Together, they create a lawful and orderly environment for project development [68].

The general public is an important actor in the realization and social communication of technological value. Public health understanding, user feedback, and public opinion can influence the decisions of policymakers and regulatory authorities [74].

Educational institutions serve as bridges for knowledge and talent development. Through science communication and health education, they improve public health literacy and provide momentum for the development of the project ecosystem.

Collectively, indirect stakeholders form a closed influence chain of "policy - regulation - context - cognition," jointly shaping social acceptance, diffusion speed, and the long-term development potential of the technology [75].

(5) Joint Advancement of Indirect and Direct Stakeholders

Centered on technology implementation and value realization, indirect stakeholders, direct stakeholders, and researchers in relevant fields form a bidirectional network of "high-level guidance - intermediate enabling - frontline implementation and feedback." The participating groups are mutually dependent, mutually influential, and jointly engaged in project advancement.

Direct stakeholders are the principal carriers of technology application and value transfer. Indirect stakeholders provide institutional support and developmental boundaries through policy constraints, regulatory standards, and public understanding, while researchers in relevant fields perform key roles in scientific validation, technological enabling, and evidence linkage, supplying theoretical foundations and technical standards across the full chain. Health-insurance subsidies and technology-promotion policies formulated by policymakers affect payer reimbursement scope and the willingness of manufacturing

partners to produce the technology. Technology-access and quality-supervision requirements imposed by regulatory authorities influence the market-entry timeline of manufacturing partners and the compliance of distributor promotion. Science communication by educational institutions can improve technology acceptance and use capability among patients, close contacts, and susceptible populations. Public perceptions and opinion can influence policy decisions and encourage direct stakeholders to improve services, while the observed effectiveness of direct-stakeholder activities can in turn shape social perceptions through word of mouth. Researchers in relevant fields provide scientific evidence for policymaking, regulatory standard-setting, and science-communication content through technology assessment, real-world studies, and feasibility evaluation, thereby strengthening the professionalism and contextual fit of high-level institutional design.

At the same time, practical feedback from direct stakeholders can drive continuous strategic refinement by indirect stakeholders and research participants. Reports from manufacturing partners and distributors on market barriers can contribute to policy optimization; application problems reported by patients and healthcare institutions can be transmitted through regulatory authorities to policymakers and support improvement of industry standards. Drawing on frontline application data and context-specific problems, researchers in relevant fields can continuously review technologies and mechanisms and provide targeted recommendations for institutional adjustment, product upgrading, and regulatory refinement, thereby linking practical problems with high-level optimization through research.

Through a complete feedback loop of "policy guidance - research enabling - implementation - feedback optimization," stakeholders can achieve close integration of high-level design, scientific research, and frontline practice, supporting compliant, efficient, and sustainable technology implementation while continuously improving social acceptance and the potential for public-health application [76].

(6) Collaborative Interconnection between Indirect and Core Stakeholders

Centered on effective technology application and the protection of health interests, indirect and core stakeholders establish a multidirectional feedback loop of "policy guidance - regulatory standardization - capacity support - demand response." This structure provides an institutional foundation for frontline diagnosis, treatment, and public health and enables iterative improvement across the full chain through practice-based feedback.

Policymakers and regulatory authorities provide the institutional foundation and interact bidirectionally with core stakeholders. Policies concerning health-insurance reimbursement and subsidies for primary-care equipment can reduce testing costs for nursing homes, kindergartens, patients, and susceptible populations while also providing financial support for equipment acquisition by healthcare institutions.

Through technology-access review and quality supervision, regulatory authorities help ensure device safety. Application problems arising in healthcare institutions, nursing homes, and kindergartens can be incorporated into regulatory feedback mechanisms, encouraging policymakers to refine standards and implementation rules and thereby forming a positive feedback cycle.

Educational institutions build bridges of understanding by using science communication to help older adults, children, patients, and susceptible populations understand testing procedures and improve

acceptance. Clinical and contextual experience can be incorporated into educational activities, while the practical experience of core stakeholders can provide material for curriculum improvement.

As a bridge for public opinion, the general public can encourage policy and regulatory improvement through its evaluations and expressed needs and can influence willingness to test among priority populations. Experiences with healthcare services, institutional testing, and patient use can spread through word of mouth and expand the social influence of the technology.

Practical feedback from core stakeholders is a major driver of strategic improvement among indirect stakeholders. Technical problems reported by healthcare institutions can promote refinement of policies and standards; prevention-and-control experience from nursing homes and kindergartens can shape public understanding and be incorporated into science education; and the needs of patients and susceptible populations can guide service upgrading and the direction of science communication.

This interaction ultimately forms a closed loop of "technology application - feedback optimization - strategy upgrading - technology implementation." With support from policy, regulation, education, and the public, testing technologies can better serve clinical care, eldercare and early-childhood settings, and public-health prevention and control, thereby contributing to the protection of population health [77].

3. Stakeholder Collaboration Strategies Based on the Power-Interest Matrix

Based on actual project operations and the specific ways in which different actors participate, the team appropriately consolidated selected stakeholder categories. Patients, susceptible populations, and close contacts were combined into a single category of patients and high-risk populations, while distributors, manufacturing partners, and health insurance payers were grouped as business partners. This consolidation avoids excessive fragmentation and more accurately reflects actual interaction patterns within the project [86]. On this basis, the team constructed a concentric-circle stakeholder map and systematically examined stakeholders' core interests, interaction relationships, and project influence according to dimensions including degree of association with the project, depth of participation, and directness of influence. The map provides a visual representation of multilateral coordination and interaction within the project ecosystem and establishes a basis for subsequent stakeholder-priority assessment and collaboration strategy design. To translate the overall relational understanding generated by the concentric-circle map into actionable management guidance, the team further introduced the Power-Interest Matrix. Stakeholders were assigned strategic priorities according to their level of power and degree of interest [89, 90], allowing the project to identify the management needs and focal concerns of different groups. Differentiated engagement and coordination strategies were then developed according to stakeholder characteristics and interests to promote demand responsiveness, resource integration, and sustained communication and to establish a stakeholder-collaboration mechanism appropriate to the project's stage of development.

3.1 High-Power, High-Interest Stakeholders

The high-power, high-interest group includes patients and high-risk populations, nursing homes and kindergartens, healthcare institutions, and public health institutions. These actors hold substantial decision-making influence over real-world project implementation and are also directly exposed to the public-health benefits and risks associated with technology use. They therefore place particular emphasis on diagnostic

performance, compliance boundaries, and long-term sustainable value. To respond to these concerns, the team establishes long-term strategic collaboration and embeds real clinical and public-health prevention needs into the research and development process from an early stage, ensuring that technological innovation remains grounded in actual application settings.

In primary-level settings, testing needs vary across congregate environments. Eldercare and childcare settings require low-threshold screening solutions that can be deployed rapidly, and product design must account for the physiological characteristics of older adults and children. Drawing on internationally established sentinel-surveillance concepts, relevant domestic regulatory frameworks, and disease-surveillance practices in long-term-care and childcare institutions in multiple regions, the team continuously optimizes the product using frontline feedback, reduces operational complexity, and improves suitability for routine screening in congregate settings [83, 88].

For patients and high-risk populations, the primary needs are convenient access to testing, clear and comprehensible operating instructions, and corresponding guidance for risk management [81, 83]. The project therefore established a full-cycle needs-feedback system combining online questionnaires, community visits, and in-person interviews. This system continuously captures practical problems associated with home testing and treats user experience as a priority in product iteration, enabling simplification of operating procedures, improvement of home-use guidance, and development of safer and more user-friendly household screening solutions.

Healthcare institutions place particular emphasis on the stability and reliability of test results and on whether testing can serve as an effective complement to conventional hospital-based services [84]. The team collaborates with healthcare professionals from respiratory medicine, laboratory medicine, infectious diseases, and related departments at tertiary hospitals. Through clinical-context research, the project iteratively optimizes testing performance and product design to develop convenient screening solutions suitable for home and eldercare settings.

For public health institutions, the technology must provide multi-pathogen surveillance data and timely risk-warning capabilities to support regional disease-control decisions. The team works with experts in disease control, epidemiology, and statistics to define three core objectives: combined multi-pathogen screening, interoperable data sharing, and simple operation in primary-level settings. Based on the epidemiological characteristics of respiratory disease in such settings, a five-target panel was selected comprising influenza, SARS-CoV-2, respiratory syncytial virus (RSV), human metapneumovirus (hMPV), and Streptococcus pneumoniae [79, 88].

Integrating these stakeholder requirements, the team links high-level design with frontline practice and develops an analytical platform connecting immune indicators with testing data. Considering age-related immune decline in older adults and the immaturity of the immune system in children, differentiated health-management strategies are formulated. The rapid-testing system is planned for deployment in community health service centers to establish an integrated chain of “nearby testing - data reporting - risk warning - targeted intervention,” thereby promoting close integration between scientific innovation and public-health practice [79, 87].

3.2 High-Interest, Low-Power Stakeholders

The high-interest, low-power group comprises business partners. These stakeholders require the project to demonstrate manufacturability at scale, effective channel distribution, and compatibility with health-insurance reimbursement so that devices can be produced in volume, deployed in primary-level settings, and made broadly accessible. The team therefore optimizes hardware structure and product design across multiple dimensions, including production scalability, channel implementation, and payment accessibility, thereby linking research and development, manufacturing, and distribution into a complete implementation pathway and supporting market-oriented and equitable translation of the device [84, 82, 86].

3.3 Low-Interest, High-Power Stakeholders

The low-interest, high-power group includes regulatory authorities and policymakers, as well as researchers in relevant fields. Regulators and policymakers hold core authority over market access, compliance, and policy formulation. Although they do not participate directly in routine project implementation, their decisions determine whether the technology can enter public settings at scale. In evaluating the project, these actors focus on compliance, controllability of risk, clinical value, and health-economic performance. The team therefore adopts a communication strategy centered on early engagement, evidence-based support, and compliance-first implementation to proactively address policy-level barriers [85].

Regulatory authorities require all stages of the project - from research and development and manufacturing to field application - to comply with ethical requirements and industry standards, with risks controlled throughout the process. The team systematically reviews relevant regulations, establishes an integrated “R&D - compliance - policy” feedback channel, regularly consults experts in ethics, public health, clinical practice, and regulation, and proactively reports experimental data, technical parameters, and implementation procedures to ensure that the project progresses within a safe and compliant framework.

Policymakers place greater emphasis on whether the technology addresses pressing public-health prevention needs and has the potential for implementation at scale. The team therefore conducts early field research in high-risk congregate settings, participates in discussions of prevention strategies and industry standards based on primary-level implementation challenges, and seeks access qualifications and policy support for priority scenarios, thereby promoting alignment between technological innovation and the public-health policy system [88].

Researchers in relevant fields require the overall testing platform to remain scientifically reliable at the molecular, algorithmic, and hardware levels while retaining realistic engineering potential. The project therefore engages these researchers in focused discussions on molecular design, computational modeling, and hardware architecture and iteratively optimizes the complete technical solution to strengthen the platform's scientific foundation [80, 81, 82, 84].

3.4 Low-Interest, Low-Power Stakeholders

The low-interest, low-power group includes educational institutions and the general public. Although these stakeholders do not directly participate in project decision-making or technical implementation, they

are essential to science communication, social understanding, and equitable diffusion of the technology. To reduce cognitive barriers surrounding synthetic biology and rapid-testing technologies, the team established an integrated communication system combining routine outreach, audience-segmented education, and interactive feedback, translating laboratory knowledge into content that the public can understand and engage with [79, 82].

Educational institutions at different levels require life-science and public-health teaching resources appropriate to different stages of education for classroom teaching and science-and-innovation activities. The team therefore develops complete sets of science-communication materials for childcare settings, primary and secondary schools, and related institutions, including illustrated materials, experimental models, traveling exhibitions, and curriculum plans. These resources can be incorporated into school science-education repositories, transforming project technology into an implementable teaching vehicle for introducing young people to the life sciences.

The general public encompasses people with widely varying levels of prior knowledge and capacity to understand new technologies and therefore requires diverse communication content adapted to different groups. The team conducts audience-segmented science communication for maternal and infant populations, older adults, adolescents, and special-education groups, designing lightweight and engaging formats around the everyday contexts of each audience while balancing scientific accuracy with human-centered communication. Feedback channels are also established to continuously collect audience experiences and suggestions, dynamically refine communication content and activity formats, and invite members of the public to participate in co-creation, thereby promoting inclusive and broadly accessible science communication.

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